Starter

Define the following words, both for regular english and as computer science terms:

1. Entity
2. Attribute

3. Relationship

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Define the following words, both for regular english and as computer science terms:

1. Entity a thing with distinct and independent existence.
2. Attribute a quality or feature regarded as a characteristic or inherent part of someone or something.
3. Relationship the way in which two or more people or things are connected, or the state of being connected.

Entities and Relationships

Keywords: Entity; Attribute; Relationship; One-to-many; many-to-many; one-to-one; data redundancy; flat-file;

Homeworks

Marks

1: Incomplete, **requires resubmission**
2: Complete, but late.
3: Complete and on time

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Missing more than one homework

- Email to you **and** your parents
- Summary note to focus. **Permenant!**
- Context based, TALK TO ME.
- May involve phonecalls.

Flat-file databases

So far, we've only looked at flat-file databases:

order number	Date	product code	desc.	quantity ordered	unit price
152117	23-5-2014	EG8766	CD player	1	59.99
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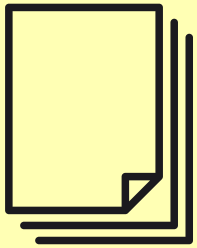
Normally, it would be better to use a **relational database** (more than one table) to store this. **Why?**

Relational Database Terms



Entity

- An object, person, event or thing of interest about which data is to be recorded
- eg: Customer, Order, Product

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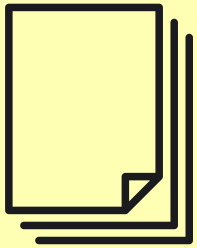

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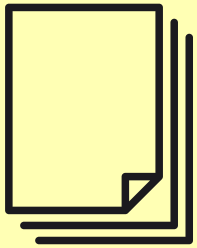

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- An object, person, event or thing of interest about which data is to be recorded
- eg: Customer, Order, Product

Attribute

- A property of an entity
- eg: Customer ID, Customer Name, Item price

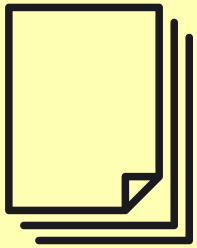
Relationship

- A link or association between two entities
- eg: A customer can make many orders, but each order has only one customer.

Relational Databases

Once decisions about the **entities, attributes and relationships** have been made, the database can be created.

Relational Databases



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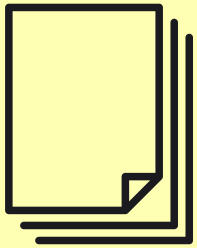

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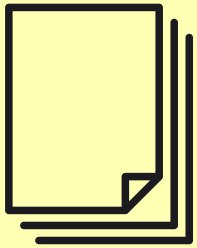

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A table is created for each entity. Each record represents a different instance of that entity.

Fields within tables correspond to attributes.

Relationships are represented with key fields that link the records together between tables.

Types of relationships

One-to-One

- Employee and Company Car;
- Husband and Husband;

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- Employee and Company Car;
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One-to-Many

- Owner and Pet;
- Borrower and Book;

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One-to-One

- Employee and Company Car;
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One-to-Many

- Owner and Pet;
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Many-to-Many

- Student and Course;
- Film and Film Star

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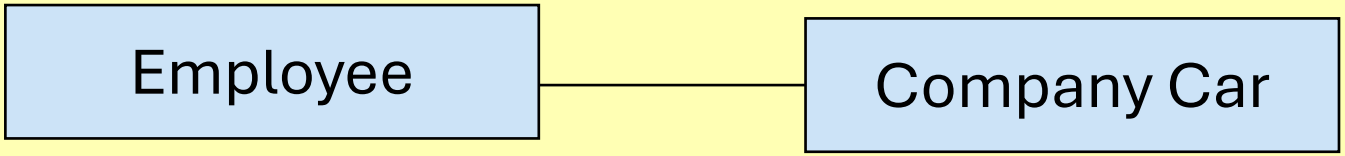

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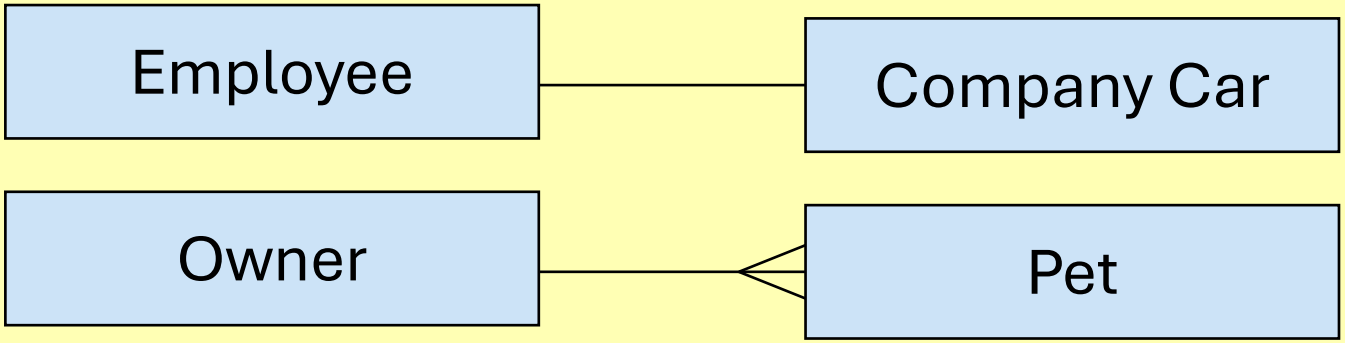

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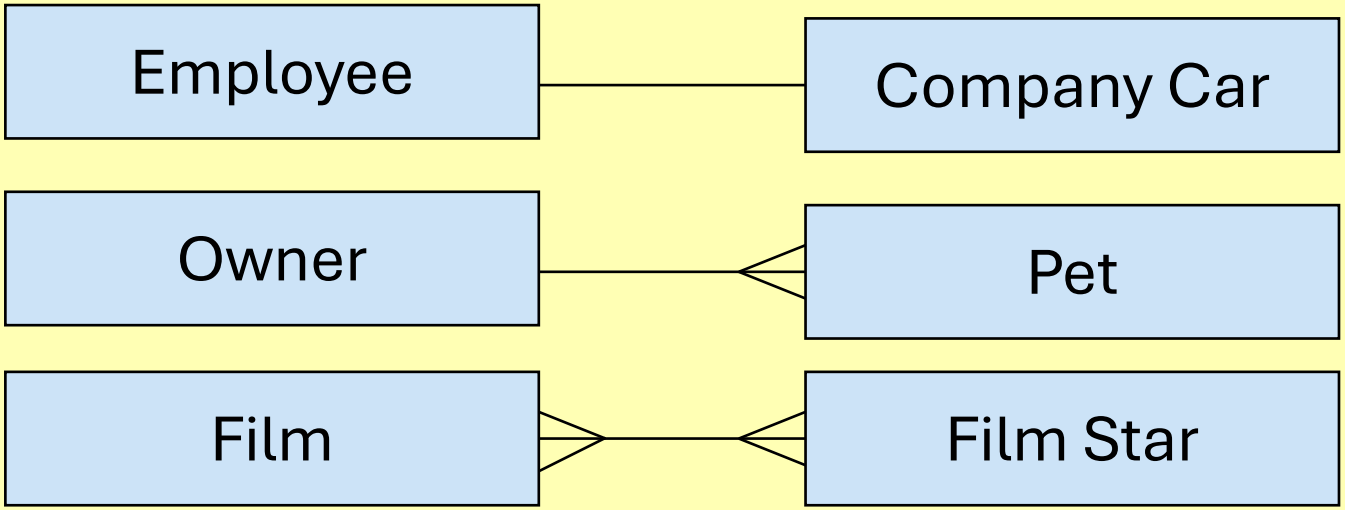


Draw the relationship!

Michael, a bookshop owner, uses a relational database to record details of books and the category in which they can be found on the shelves. A unique International Standard Book Number (ISBN) is assigned to each book. Each book can be allocated one category, e.g. fiction, home and garden, reference, etc. Each category can have many books. What is the relationship between the Category and the Book?

Draw the relationship!

Sophie is a printmaker who specialises in limited edition prints of dogs that she creates from her original photographs. She creates 100 identical prints from each photograph. She sells her prints through a gallery in her local town. Each photograph is assigned a unique name. She records the name of the photograph each customer has bought but does not record the details about the print they buy. Her sales are stored in a database with three entities: Photograph, Sales and Customers. What is the relationship between the Customer and the Photograph?

Draw the relationship!

Bradley, an A-level Computer Science student, has started a blog to share resources on his computer science lessons with his fellow students. He posts to his blog after each lesson a summary of the theory covered that day. His fellow students can then add their comments to each post. His blog is stored in an online database with two entities – BlogPost and Comment. What is the relationship between BlogPost and the Comment?

Draw the relationship!

Louise is taking an evening class in French. She has decided to record her French vocabulary in a database so that she can easily search for words as she needs them. Her French teacher has given her 100 pairs of English-French translations to learn. She has stored the French words in a table and the English words in a separate table. What is the relationship between the French words and the English words?

Let’s design

Database entities for a car repair shop

- Customers: stores the customers ID number, name, address, phone number
- Cars: store the number plate, make, model, colour, owner
- Employee: stores the employee’s ID number, name, address, phone number, job title, salary
- Job: stores the Job ID, the car reg, date, time and the ID of the employee working on that job

You Try:

Database entities for a zoo

- The database needs the information about the animals they keep, they need the name, species, date of birth and international zoo identification tag.
- The database needs to store information about the employees – it needs their employee IDs, names, addresses, email addresses, start date and number of days per week they work
- There will also be information about the enclosures; each enclosure has a name, a size in metres squared, opening and closing times
- Each enclosure will contain some animals and each employee will be responsible for the upkeep of only one enclosure. Some enclosures are large and need multiple employees.

Create 3 tables and then draw an E-R diagram

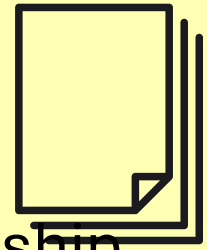
Modification

Now, the database needs to be modified so that each employee is responsible for the upkeep of **some enclosures**, rather than just one.

Some enclosures still require several employees to look after.

What's the problem now?

Linking Tables



By using foreign keys, we can create a one-to-many relationship within the database.

In order to create a **many to many** relationship, we need a **linking table**.

EG: Students and Courses

From earlier

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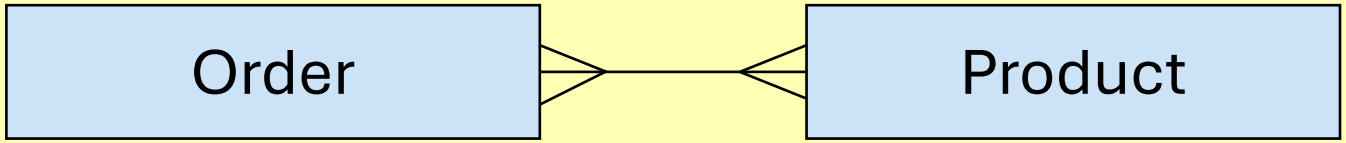
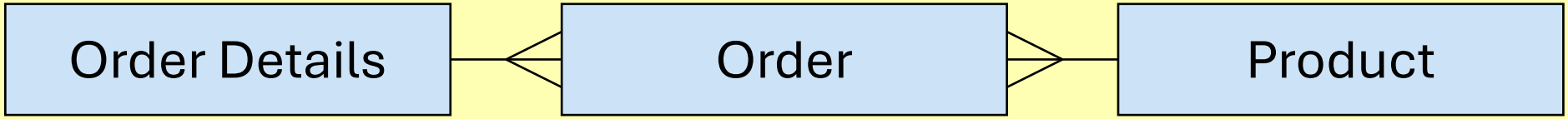
1. What would be two sensible entities to break this flat file into?
2. Draw an ER diagram
3. Write the table declaration statements.

Solution:

1. Order and Product
2. 

Alternatively:



3.

Order (<u>OrderEntryNumber</u>, OrderNumber, ProductCode, Quantity, Date)
Product (<u>ProductCode</u>, Description, UnitPrice)

Solution:

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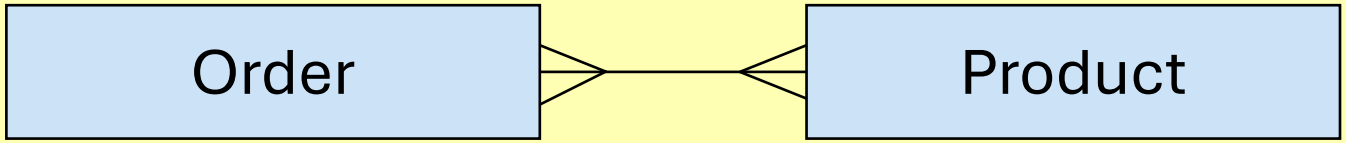
2. 

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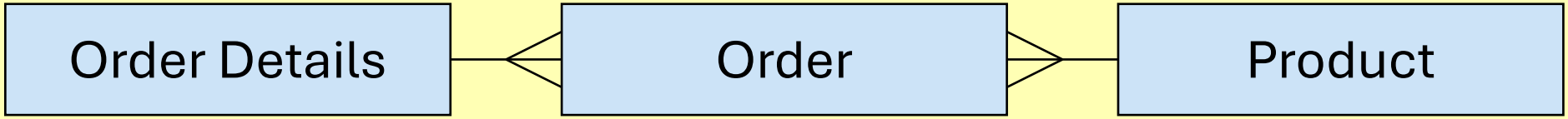

3. Alternatively:

   Order Details (OrderNumber, Date)

   Order (OrderNumber*, ProductCode*, Quantity)

   Product (ProductCode, Description, UnitPrice)

Why might we want that more-complex set of tables?

Complex:

Order Details (OrderNumber, Date)

Order (OrderNumber*, ProductCode*, Quantity)

Product (ProductCode, Description, UnitPrice)

Simpler:

Order (OrderEntryNumber, OrderNumber, ProductCode, Quantity, Date)

Product (ProductCode, Description, UnitPrice)

You Try

Take the information in this flat-file database:

Customer Number	Customer Name	Company Name	Town	Postcode	Car Reg	Make	Model	Hire Date	Return Date
153	Narazali, N	MF Plastics	Derby	DB5 3ER	KL52 HGT	Ford	Focus	12.2.14	25.3.14
153	Narazali, N	MF Plastics	Derby	DB5 3ER	HV53 SWR	Ford	Focus	1.4.14	22.5.14
187	Brown, L	MF Plastics	Derby	DB5 3ER	HX03 TYR	Nissan	Qashqai	2.4.14	12.4.14
287	Pinner, M	Tooks Ltd	Derby	DB6 8YU	KL52 HGT	Ford	Focus	3.1.14	13.1.14
287	Pinner, M			DB6 8YU	HJ04 FGT	Honda	Jazz	15.3.14	25.3.14

1. What are the entities?
2. Draw an ER diagram
3. Which fields/attributes would be in each entity?
4. Have you removed **all** data redundancy? If not, can you add more entities to help?

Extension: Write the DDL to create the tables.